

Formula

1. WB + WA

$$\text{Accesses via the BUS} = \text{freq} \left[\frac{\text{words}}{\text{sec}} \right] \cdot \text{MR} \cdot (\% \text{ dirty} \cdot \text{WMP}_{\text{WB}}^b + \% \text{ clean} \cdot \text{RMP}_{\text{WB}}^b)$$

$$\text{Write Miss Penalty}_{\text{WB}}^b = \text{Read Miss Penalty}_{\text{WB}}^b$$

$$\text{WMP}_{\text{WB}}^b = \underbrace{\frac{\text{block size}}{\text{bus width}} \cdot \text{BUS Reads}}_{\text{ALLOCATE}} + \underbrace{\frac{\text{block size}}{\text{bus width}} \cdot \text{dirty} \cdot \text{BUS writes}}_{\text{UPDATE}}$$

2. WB + WNA

$$\text{Accesses via the BUS} = \text{freq} \left[\frac{\text{words}}{\text{sec}} \right] \cdot \text{MR} \cdot (\% \text{ dirty} \cdot \text{WMP}_{\text{WB}}^b + \% \text{ clean} \cdot \text{RMP}_{\text{WB}}^b)$$

$$\text{RMP}_{\text{WB}}^b = \underbrace{\frac{\text{block size}}{\text{bus width}} \cdot \text{BUS Reads}}_{\text{ALLOCATE}} + \underbrace{\frac{\text{block size}}{\text{bus width}} \cdot \text{dirty} \cdot \text{BUS writes}}_{\text{UPDATE}}$$

$$\text{WMP}_{\text{WB}}^b = 1$$

3. WT + WA

$$\text{Accesses via the BUS} = \text{freq} \left[\frac{\text{words}}{\text{sec}} \right] \cdot \text{MR} \cdot (\% \text{ dirty} \cdot \text{WMP}_{\text{WT}}^b + \% \text{ clean} \cdot \text{RMP}_{\text{WT}}^b) +$$

$$\underbrace{\text{pt. dirty} \cdot \text{RMP}_{\text{WT}}^b}_{\text{pt. dirty} \cdot \text{RMP}_{\text{WT}}^b} + \text{freq} (1 - \text{MR}) \cdot \text{WHP}_{\text{WT}}^b \cdot \% \text{ dirty}$$

$$\text{RMP}_{\text{WT}}^b = \frac{\text{block size}}{\text{bus width}} \text{BR}$$

$$\text{WMP}_{\text{WT}}^b = \frac{\text{block size}}{\text{bus width}} \text{BR} + 1 \text{ BW}$$

$$\text{WHP}_{\text{WT}}^b = 1 \text{ BW}$$

4. WT + WNA

$$\text{Accesses via the BUS} = \text{freq} \left[\frac{\text{words}}{\text{sec}} \right] \cdot \text{MR} \cdot (\% \text{ dirty} \cdot \text{WMP}_{\text{WT}}^b + \% \text{ clean} \cdot \text{RMP}_{\text{WT}}^b) +$$

$$\underbrace{\text{pt. dirty} \cdot \text{RMP}_{\text{WT}}^b}_{\text{pt. dirty} \cdot \text{RMP}_{\text{WT}}^b} + \text{freq} (1 - \text{MR}) \cdot \text{WHP}_{\text{WT}}^b \cdot \% \text{ dirty}$$

$$\text{RMP}_{\text{WT}}^b = \frac{\text{block size}}{\text{bus width}}$$

$$\text{WMP}_{\text{WT}}^b = 1 \text{ BUS Write}$$

$$\text{WHP}_{\text{WT}}^b = 1 \text{ BUS Write}$$

Ex. 1

Un sistem de calcul are un procesor ce trimite referințe către memoria cache la o frecvență de 10^8 words/second, unde 10% sunt scrieri, iar 90% citiri.

Cache-ul are un miss rate de 10% și un bloc de 8 cuvinte.

Bus-ul are un bandwidth de 10^9 accese/secundă și o lățime (bus bandwidth) de 1 cuvânt.

La orice moment, 20% din blocuri sunt dirty.

Să se calculeze procentajul din bus bandwidth folosit din cauza imperfecțiunii cache-ului în urm. cazuri: $A_2, B_1 \rightarrow$ date diferite

A1

a) $WB + WA$

b) $WT + WNA$

a)

$$\text{Accesses via the BUS} = \text{freq} \left[\frac{\text{words}}{\text{sec}} \right] \cdot \text{MR} \cdot (\% \text{ scrieri} \cdot \text{WMP}_{WB}^b + \% \text{ citiri} \cdot \text{RMP}_{WB}^b)$$

$$\text{WMP}_{WB}^b = \frac{\text{block size}}{\text{BUS width}} \cdot \text{BUS Reads} + \frac{\text{block size}}{\text{BUS width}} \cdot \text{dirty} \cdot \text{BUS writes} =$$

ALLOCATE UPDATE

$$= \frac{8}{1} + 0,2 \cdot \frac{8}{1} = 1,2 \cdot 8 = 9,6 \text{ BUS Accesses}$$

$$\text{AVTB} = 10^8 \cdot 0,1 \cdot (0,1 \cdot 9,6 + 0,9 \cdot 9,6) = 10^7 \cdot 9,6 \text{ BUS Accesses}$$

$$\% \text{ Used BUS}_{WB+WA} = \frac{\text{AVTB}}{\text{BUS bandwidth}} = \frac{10^7 \cdot 9,6}{10^9} = 9,6\%$$

b)

$$\text{Accesses via the BUS} = \text{freq} \left[\frac{\text{words}}{\text{sec}} \right] \cdot \text{MR} \cdot (\% \text{ scrieri} \cdot \text{WMP}_{WT}^b + \% \text{ citiri} \cdot \text{RMP}_{WT}^b) +$$

pt. hit P.

$$\rightarrow + \text{freq} (1 - \text{MR}) \cdot \text{WHP}_{WT}^b \cdot \% \text{ scrieri}$$

$$\text{RMP}_{WT}^b = \frac{\text{block size}}{\text{bus width}} = \frac{8}{1} = 8 \text{ BUS Reads}$$

$$\text{WMP}_{WT}^b = 1 \text{ BUS Write}$$

$$WMP_{WT}^b = 1 \text{ BUS Write}$$

$$AVTB = 10^8 \cdot 0,1 \cdot (0,1 \cdot 1 + 0,9 \cdot 8) + 10^8 \cdot 0,9 \cdot 1 \cdot 0,1 = \\ = 10^7 \cdot (7,3 + 0,9) = 8,2 \cdot 10^7$$

$$\% \text{ Used BUS }_{WT+WNA} = \frac{8,2 \cdot 10^7}{10^9} = 8,2\%$$

B₂ am schimbător : a) WB + WNA b) WT + WNA

a)

$$\text{Accesses via the BUS} = \text{freq} \left[\frac{\text{words}}{\text{sec}} \right] \cdot MR \cdot (\% \text{ scrieri} \cdot WMP_{WB}^b + \% \text{ citiri} \cdot RMP_{WB}^b)$$

$$RMP_{WB}^b = \frac{\text{block size}}{\text{BUS width}} \cdot \text{dirty BUS Writes} =$$

$$WMP_{WB} = 1$$

Normalize

Exercise 5

A processor sends references to the memory with frequency 10^{10} words/sec (10% writes (A)/15% writes (B)). The cache has a miss rate of 8% (A)/ 12% (B) and a block of 16 words (A)/4 words (B), with 20% (A)/14%(B) being dirty at any time. The bus has a width of 4 words and a capacity of $2 \cdot 10^{10}$ accesses/sec (A,B). Calculate the percentage of BUS capacity used because of the cache imperfection if the write policy is:

- a) write back + write allocate
- b) write through + write no allocate

5. Un procesor trimite referințe unei memorii cu o frecvență 10^{10} words/sec (10% writes). Cache are miss rate de 8% și un block de 16 words, cu 20% dirty at any time. The bus are width de 4 words și capacitate de $2 \cdot 10^{10}$ accesses/sec. Calculați procentajul BUS capacity of the cache implementation if the write policy is:

- a) write back + write allocate
- b) write through + write no allocate